

Contrastive Alignment of Text and Vital Signs

Author Name

June 1, 2026

1 Contrastive Alignment: Sharing a Clinical Embedding Space

1.1 The Core Concept: A Shared Room

BioBERT maps clinical text into a high-dimensional vector space. In this setting, tokenized symptom descriptions are converted into dense contextual embeddings with dimensionality

$$d = 768.$$

One can imagine this 768-dimensional space as a large room. BioBERT may place the phrase “my chest is extremely tight” at some location in this room. However, without additional training, BioBERT is blind to physiological measurements: it does not inherently know what a heart rate of 130 beats per minute means, nor how that vital sign should relate to the symptom description.

Contrastive alignment addresses this limitation by forcing both text embeddings and TEWS vital-sign embeddings into the same representational space. During training, matching text–vitals pairs are pulled closer together, while mismatched pairs are pushed farther apart.

1.2 The Twin Network Architecture

The model is trained using two encoders side by side:

- **Network A: Text Encoder.** The existing BioBERT model takes the text input X and outputs a 768-dimensional vector E_{text} .
- **Network B: Vitals Encoder.** A small multilayer perceptron (MLP) takes the numerical TEWS array, such as heart rate, respiratory rate, temperature, and AVPU score, and maps it through several neural layers until it also outputs a 768-dimensional vector E_{vitals} .

The goal is for both networks to describe the same patient state using vectors that live in the same embedding space.

1.3 The Mathematics of Push and Pull

Once both encoders produce their vectors, the system measures their closeness using cosine similarity:

$$\text{similarity}(E_{\text{text}}, E_{\text{vitals}}) = \frac{E_{\text{text}} \cdot E_{\text{vitals}}}{\|E_{\text{text}}\| \|E_{\text{vitals}}\|}.$$

The actual learning is driven by a contrastive loss function, commonly InfoNCE loss. During training, the model receives batches of patients and applies two complementary rules:

- **The Pull: Positive Pairs.** If E_{text} represents a phrase such as “I can’t catch my breath” and E_{vitals} represents the same patient’s vital signs, such as respiratory rate ≥ 26 , the loss penalizes the model when the vectors are far apart. This encourages both encoders to update their weights so the paired vectors move closer together.
- **The Push: Negative Pairs.** The model also compares that same text embedding against the vitals embedding of a different patient in the batch, for example a lower-acuity patient with normal breathing. The loss repels these mismatched vectors, pushing them farther apart in the shared embedding space.

In effect, the system learns not only which text and vitals belong together, but also which combinations are clinically inconsistent.

1.4 Why This Is Powerful at Inference Time

The most useful part of this approach is that the vitals encoder can be discarded after training. When the web application is deployed, the patient may only provide free-text symptoms, and vital signs may not yet have been measured.

However, because BioBERT was mathematically forced during training to align its text embeddings with real physiological measurements, its representations become anchored to clinical reality. If a patient writes something vague, such as “I feel like an elephant is sitting on me,” the text encoder can map that phrase into a region of the embedding space associated during training with severe clinical patterns, such as tachycardia, respiratory distress, or hypoxia.

This gives the text model a form of learned clinical intuition: even without receiving vitals at inference time, it has learned how language tends to correspond to physiological severity.

1.5 Visual Interpretation

Conceptually, contrastive training reorganizes the model’s embedding space. Before alignment, symptom phrases and vital-sign patterns may occupy unrelated regions. After alignment, clinically

matching text and vitals cluster together, while unrelated or inconsistent pairs are separated. This push-and-pull process reshapes the model’s internal representation so that language becomes more tightly connected to patient physiology.